CTS — Aptitude Test 3

Questions 51–75 of 100 · 25 questions

Question 51. Increasing of length and breadth is proportional. If length 6m is changed to 21m and breadth changes to 14m then what was the previous value of breadth?

Solution:
Speed of first train = (350/2) kmph

= 175 kmph.
Ratio between the speed of two trains
= 5 : 3
Speed of train A : Speed of train B
= 175 : X
5 : 3 = 175 : X
X = 105 kmph.

Solution:
Length1 : Breadth1 :: Length2 : Breadth2
6 : B1 :: 21 : 14
6/B1 = 21/14
B1 = 6x14/21
B1 = 4m.

Question 52. Ramesh, xyz and Rajeev put a partnership. Total profit is 36000. If Ramesh and xyz invested in the ratio 5:4 and xyz and Rajeev invested in the ratio 8:9, find Rajeev's share.

Solution:
Ratio of Ramesh and xyz = 5 : 4 = 10 : 8
Ratio of xyz and Rajeev = 8 : 9
Then ratio of Ramesh : xyz : Rajeev is
= 10 : 8 : 9
Share of Rajeev=36000 x [9/(10+8+9)]
= 36000 x (1/3)
= 12000

Question 53. A Product is supported each week by the same three Customer Service Representatives (CSR's). Last month the first CSR took 450 calls, the second took 350 calls, and the third took 300 calls. This month the job will consists of 1500 calls. If the three CSR's each increase

their work proportionately, how many more calls will the first CSR take this month than last month?

Solution:
The number of calls for first CSR out of total 1100 calls = 450.
Since the number of calls increase proportionally, the number of calls attended by him in this month will be:
450/1100 = X/1500
X = 450 x 1500/1100
X ■613 calls.

Question 54. Diamonds value is proportional to square of its weight. When diamonds broke into pieces in ratio 1:2:3:4:5 total loss in value is 85000. What is the value of the diamond twice the weight of original diamond?

Solution:
Let the actual weight of the diamond pieces after broken down be
1x, 2x, 3x, 4x, 5x.
Total weight of the diamonds = 15x
Value of the diamond as a whole
= (15x)2 = 225x

Value of each piece after broke down
= (1x)2 + (2x)2 + (3x)2 + (4x)2 + (5x)2

= x2 + 4x2 + 9x2 + 16x2 + 25x2

= 55x2

Difference between the two values
= 225x2 – 55x2 = 85000
= 170x2 = 85000
x2 = 500
Actual weight = 15x
Double the weight = 30x
Price of a diamond with weight 30x
= (30x)2

= 900x2

= 900 x 500
= 450000

Question 55. Every year before the festive season, a shopkeeper increases the price of the product by 35% and then introduce two successive discount of 10% and 15% respectively. What is percentage loss or percentage gain through this transaction?

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Solution:
Assume that the cost price of the product is Rs. 100.
After marking it up by 35%, the marked price will be = Rs. 135
10% discount from 135 will give
135 – (10/100)135 = 121.5
From 121.5 after giving 15% discount we get,
121.5 – (15/100)121.5
= 103.275
The selling price is more than cost price.
The profit percentage is
= [(103.275 – 100)/100]100
= 3.275% profit.

Question 56. Steward assign 1/8th of his monthly salary for food. Steward's total food bill for the month is Rs.6500. What is Steward's yearly salary?

Solution:
(1/8) x Monthly salary = 6500
Monthly salary = 6500 x 8 = 52000
Annual salary = 52000 x 12
= 624000

Profit earned = 10(5/14) = (145/14)%
Selling price of the product
= [100 + (145/14)]% of Cost price
= (1545/14)% of CP
Selling price = [(1545/14)/100] x 280
Selling price = Rs. 309
Selling price is 25% less than

the
marked price.
S.P = M.P – (25/100)M.P
S.P = (75/100)M.P
M.P = 309 x 100/75
Marked price = Rs. 412

Question 57. Manu has invested 30% of the capital in Petro bonds and rest in LIC plan. He has invested Rs.34000 more in LIC plan than in Petro bonds. How much is the total investment made by Manu?

Solution:
Investment % made in LIC = 100 – 30
= 70%
Percentage of amount invested in LIC is
70 – 30 = 40% more than Petro bonds.
40% of total investment = 34000
So,
[(40/100)T.I] = 34000
Total Investment = 34000 x 100/40
= Rs. 85000

Question 58. A supplier supplies cartridges to a news paper publishing house. He earns a profit of 20% by selling cartridges for Rs. 540. Find the cost price of the cartridges?

Solution:
To get 75% profit by selling at Rs. 75, the cost price of the spirit must be
S.P = (175/100)C.P
C.P = 75 x 100/175
C.P = 300/7
By using allegation rule, we can find out the ratio between spirit and water.

Solution:
Selling Price = 120% of Cost Price
540 = (120/100)C.P
Cost price = 540 x 100/120
Cost price = 450

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Question 59. Ram sold his car for RS. 50,000 less than what he bought it for and lost 8%. At what price should he have sold the car, if he wanted to gain as much as he lost in the first transaction?

The ratio between spirit and water
= 5 : 2

Solution:
Loss = 8% of Cost price
50000 = (8/100)CP
CP = 625000
To gain 8%, he should sell at a profit of
Rs. 50000.
Selling price at 8% gain = 675000

Question 60. A man sells his articles in such a way that even after allowing 25% discount on cash purchase, he gains 10 5/14 %. If the cost price of the articles is Rs.280 then the labeled price is?

Solution:

420/7
0

300/7

300/7
120/7
:

Question 61. A man buys a spirit at Rs. 60/lt. and adds water to it and then sells it at Rs. 75/lt. What is the ratio of spirit to water if his profit in the deal is 75%?

Question 62. Buy one get 1 free offer. Selling price of a t-shirt is 4200. Shopkeeper says he got 33.33% profit. What is cost price?

Solution:
Shopkeeper sold 2 t-shirts for Rs. 4200
as one is free.
So, selling price of one t-shirt = Rs. 2100
S.P = C.P + (33.33/100)C.P
S.P = (133.33/100)CP
CP = 2100 x 100/133.33
C.P = 1575
Cost price of 1 t-shirt = Rs. 1575

Question 63. At flat 40% discount a girl buys one jacket for Rs. 480. What is the marked price?

Solution:
SP = MP – (40/100)MP
480 = (60/100)MP
MP = 480 x 100/60
Marked price = Rs. 800

Question 64. Find out Difference between SI and CI if p=1000000 r=4% ant t=3 years?

Solution:
Simple interest for 3 years
= 1000000 x 3 x 4/100
= 120000
Compound interest for first year
= 1000000 x 4/100
= 40000
Compound interest for second year
= 1040000 x 4/100
= 41600
Compound interest for third year
= (1000000+40000+41600) x 4/100
= 1081600 x 4/100
= 43264
Total compound interest
= 124864
Difference = 120000 – 124864
= Rs. 4864

Question 65. Shopkeeper bought 400 meter cloths at Rs. 40,000 and sells at Rs. 200 per one and half meter cloth. What is the gain or loss percent?

Solution:
Cost price of 1 meter cloth = 40000/400
= Rs. 100
Cost price of 1.50 meter cloth = Rs. 150
Selling price of 1.50 m cloth = Rs. 200
Profit % = [(S.P – C.P)/C.P] x 100
= [(200-150)/150] x 100
= 33.33%

Question 66. A walks 20m towards north-east. B walk towards east 8m and then 12m south from the same point as A. Now calculate the distance between A and B?

Solution:
The
following
figure
represents
the
above question.

A

8m

D

12m

B
C

The dark lines are the path travelled by
A and B.
We have to find the distance of the

dotted line AB.
Since A travelled 20m in
North-east
direction, the distance covered by him
can be considered as the diagonal of a
square.
AD is the side of that square.
Side = Diagonal/√2
AD = 20/√2 = 10√2
BC = 10√2 – 8
AC = 10√2 + 12
ABC is a right angled triangle with AB as
the hypotenuse side.
AB2 = AC2 + BC2

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AB2 = (10√2 + 12)2 + (10√2 – 8)2
AB2 =(200+144+240√2)+(200+64-160√2)
AB2 = 608 - 80√2
AB2 = 608 – 113.2 = 494.8
AB = 22.24

Question 67. From a circular sheet of paper of radius 10 cm, a sector of area 40% is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and the height of the cone is?

Solution:
Circumference of the base of the cone
= 60% of circumference of the circle.
Circumference of circle
= 2πr = 2π(10)

Question 68. It takes 52 days to complete an agreement deal by a certain number of men. After 17 days 300 men are added and 21 days are reduced. how many men were working initially?

C = 24
C alone can do the work in 24 days.

Question 69. A, B and C together can complete a work in 8 days. All the 3 started the work together but C quit after 2 days. If the remaining work is now completed by A and B in 9 days, then how many days will C alone will take to complete the total work?

Circumference of base of the cone
= (60/100)2π(10)= 2π(6)
Radius of base of the cone:
2πrc = 2π(6)

rc = 6
The
following
image
gives
the
description of the cone thus formed.

Solution;
One day work of A, B and C = 1/8
Work done by A, B and C in two days
= 2 x (1/8) = ¼
Remaining work = 1 – ¼ = ¾
3/4th of the work is done by A and B in 9
days.
Time taken for A and B to complete full
work = 9 x 4/3 = 12 days.
One day work of A and B = 1/12

h
r

rc

r, rc and h form a right angled triangle.
r2 = rc

2 + h2

102 = 62 + h2

h = 8cm
Ratio between radius and height of the
cone is
6 : 8 = 3 : 4

C T S

Question 70. 4 men can repair a road in 7 hours. How many men are required to repair the road in 2 hours?

Solution:
After 17 days of work, out of 52 days, the
remaining number of days required is
35.
The amount of work to be done by N
number of men or N+300 number of
men remains the same.
Work done = Resource x Time
Work done by N men = N x 35
Work done by N+300 men
= (N+300)x14
So,
35N = 14N + 4200
21N = 4200
N = 200
200 men were working initially.

Solution:
Resource is inversely proportional to

time.

Let X be the number of men required to complete the work in 2 hours.
Then,

X = 14
14 men are required.

C
+
=
1

1

1

B
+
1

8

A

1

1

B
+
=
1

A

12

1

1

1

=
+

8

C

12

1
24
1

=

C

T2
T1

R1
R2

=

2

4

=

7

X

Question 71. A large rubber cushion can be filled with air pump in 10 minutes, another pump can fill the same cushion in 12 minutes. If both the pumps operate together, how long will it take to fill the cushion?

Solution:
The formula to find out time taken to
complete
the
work
together
by
2
individuals (T) is:

1

1

B
=
+
1

A

T

1

1

12
=
+
1

T

10

T = 60/11 minutes.

Question 72. A is thrice as good a workman as B and takes 10 days less to do a piece of work than B takes. the number of days taken by B to finish the work is?

Solution:
Work done by B in one day = 1/B
Work done by A in one day = 3/B
Time taken by A to complete the work is

= B/3
Difference in tie taken by A and B to complete the work
= B – (B/3) = 10
= 2B/3 = 10
B = 15 days.
B alone takes 15 days to complete the work, whereas A takes only 5 days.

Question 73. Two Copiers are being used to produce 2400 copies of a 1-page document. One Copier runs at 150% of the speed of the other. How many copies should be made on the faster copier so that both copiers will finish the same time?

Solution:
When one copier prints 100 copies, the other prints 150 copies in the same time.
Ratio between number of copies in each

printer is 2 : 3.
By dividing the total number of copies (2400) in the ratio 2 : 3, we get,
Number of copies by the slower machine
= (2/5)2400 = 960 copies
Number of copies by the faster machine
= (3/5)2400 = 1440

Question 74. A can construct a wall in 40 minutes and B can construct the wall in 45 minutes. How many hours is needed to construct a wall if both the persons are working together?

Solution:
The formula to find out time taken to complete the work together by 2 individuals (T) is:

1

1

B
=
+
1

A

T

1

1

45
=
+
1

40

T

C T S

T = 360/17 hours

Question 75. A and B can do a piece of work in 28 days. With the help of C, they can finish it in 21 days . How long will C alone take to finish the work?

Solution:

1

B
=
+
1

1

1

C
+

---(1)

A

21

1

1

B
=
+
1

---(2)

A

28

Substitute (2) in (1), we get,

1

1

C
=
+
1

28

21

C = 12 days.